Microeconomics
Course Code: Hum 2101

Lecture 7: Consumer Behavior Theory

Consumption Choices

The choices that you make as a buyer of goods and services—your consumption choices—are influenced by many factors. We can summarize them into two broad headings:

- Consumption possibilities
- Preferences

Consumption Possibilities

Your consumption possibilities are all the things that you can afford to buy. You can afford many different combinations of goods and services, but they are all limited by your income and by the prices that you must pay.

For example, you might decide to spend a big part of your income on a gym membership and personal trainer and little on movies and music, or you might spend lots on movies and music and use the free gym at school.

The easiest way to describe consumption possibilities is to consider a model consumer who buys only two items. That’s what we’ll now do. We’ll study the consumption possibilities of Lisa, who buys only movies and soda.

Consumption Possibilities

A Consumer's Budget Line

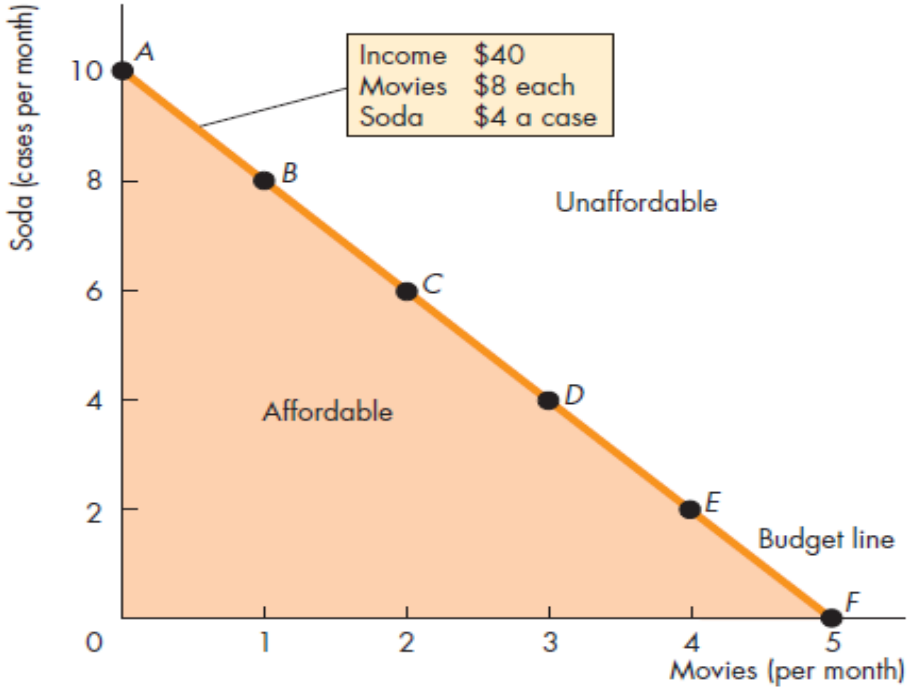
Consumption possibilities are limited by income and by the prices of movies and soda. When Lisa spends all her income, she reaches the limits of her consumption possibilities. We describe this limit with a ***budget line***, which marks the boundary between those combinations of goods and services that a household can afford to buy and those that it cannot afford.

Consumption Possibilities

Possibility	Movies Quantity	Movies Expenditure (dollars)	Soda Cases	Soda Expenditure (dollars)
A	0	0	10	40
B	1	8	8	32
C	**2**	**16**	**6**	**24**
D	3	24	4	16
E	4	32	2	8
F	5	40	0	0

The graph and the table show six possible ways in which Lisa can allocate $40 to movies and soda. In row *C* and at point *C*, she sees 2 movies and buys 6 cases of soda. The line *AF* is Lisa's budget line and is a boundary between what she can afford and what she cannot afford. Her choices must lie along the line *AF* or inside the orange area.

FIGURE 8.1 Lisa's Budget Line



Consumption Possibilities

This figure shows Lisa's consumption possibilities of movies and soda and her budget line.

Lisa has an income of $40 a month, the price of a movie is $8, and the price of soda is $4 a case.

Rows A through F in the table show six possible ways of allocating $40 to these two goods.

For example, in row A Lisa buys 10 cases of soda and sees no movies; in row F she sees 5 movies and buys no soda; in row C she sees 2 movies and buys 6 cases of soda. Points A through F in the graph illustrate the possibilities presented in the table, and the line passing through these points is Lisa's budget line.

The budget line constrains choices: It marks the boundary between what is affordable and unaffordable. Lisa can afford all the points on the budget line and inside it. Points outside the line are unaffordable.

Consumption Possibilities

Changes in Consumption Possibilities

Consumption possibilities change when ***income or prices change***.

A rise in income shifts the budget line outward. Our goal is to predict the effects of such changes on consumption choices. To do so, we must determine the choice a consumer makes.

The budget line shows what is possible; preferences determine which possibility is chosen. We'll now describe a consumer's preferences.

Preferences

Lisa's income and the prices that she faces limit her consumption choices, but she still has lots of choices. The choice that she makes depends on her preferences—a description of her likes and dislikes.

The goal of a theory of consumer choice is to derive the demand curve from a deeper account of how consumers make their buying plans. That is, we want to explain what determines demand and marginal benefit.

To achieve this goal, we need a deeper way of describing preferences. One approach to this problem uses the idea of utility, and *defines utility as the benefit or satisfaction that a person gets from the consumption of goods and services.* We distinguish two utility concepts:

- Total utility
- Marginal utility

Preferences

Total Utility

The total benefit that a person gets from the consumption of all the different goods and services is called total utility.

Total utility depends on the level of consumption—more consumption generally gives more total utility.

To illustrate the concept of total utility, think about Lisa's choices.

We give her two starting points:

(1) We will call the total utility from no movies and no soda zero utility;

(2) We will call the total utility she gets from seeing 1 movie a month 50 units.

We then asked Lisa to tell us, using the same scale, how much she would like 2 movies, and more, up to 10 movies a month. We also asked her to tell us, on the same scale, how much she would like 1 case of soda a month, 2 cases, and more, up to 10 cases a month.

Preferences

Marginal Utility

The expression “marginal” is a key term in economics and always means “additional” or “extra.” Marginal utility denotes the additional utility you get from the consumption of an additional unit of a commodity.

One of the fundamental ideas behind demand theory is the law of diminishing marginal utility. This law states that the amount of extra or marginal utility declines as a person consumes more and more of a good. To understand this law, first, remember that utility tends to increase as you consume more of a good. However, as you consume more and more, your total utility will grow at a slower and slower rate. This is the same thing as saying that your marginal utility (the extra utility added by the last unit consumed of a good) diminishes as more of a good is consumed.

The law of diminishing marginal utility states that, as the amount of a good consumed increases, the marginal utility of that good tends to decline.

Movies			Soda		
Quantity (per month)	Total utility	Marginal utility	Cases (per month)	Total utility	Marginal utility
0	0	 50	0	0	 75
1	50	 40	1	75	 48
2	90	 32	2	123	 36
3	122	 28	3	159	 24
4	150	 26	4	183	 22
5	176	 24	5	205	 20
6	200	 22	6	225	 13
7	222	 20	7	238	 10
8	242	 17	8	248	 7
9	259	 16	9	255	 5
10	275		10	260	

Preferences

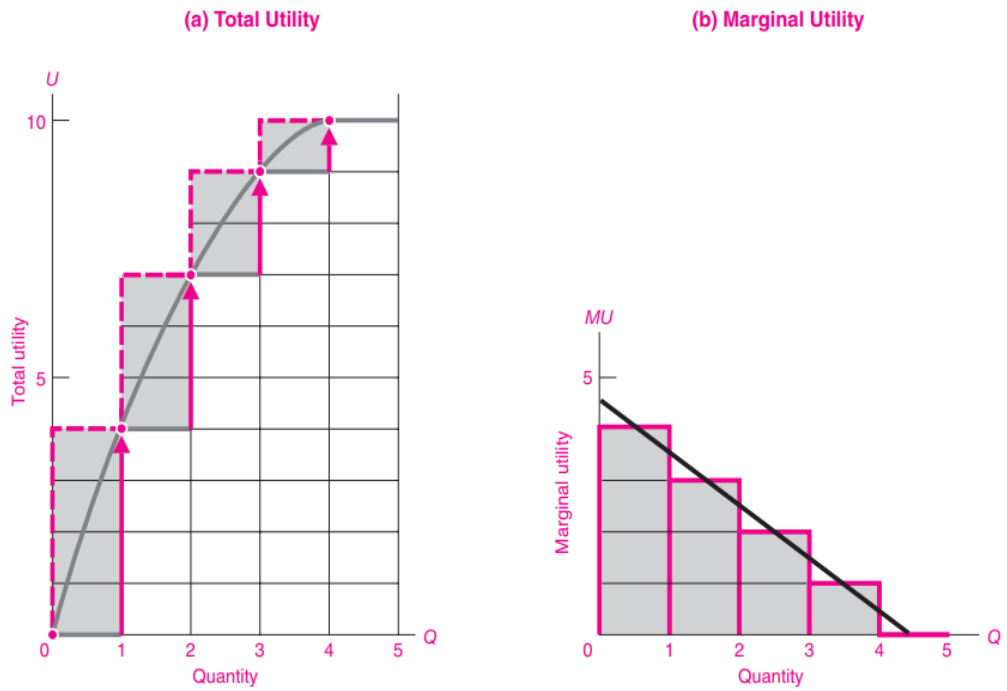
Total Utility and Marginal Utility

- Looking at these numbers, you can say a lot about how much Lisa likes soda and movies.
- She says that 1 case of soda gives her 75 units of utility—50 percent more than the utility that she gets from seeing 1 movie.
- You can also see that her total utility from soda climbs more slowly than her total utility from movies.

- "Marginal utility" column shows Lisa's marginal utility from movies and soda. You can see that if Lisa increases the soda she buys from 1 to 2 cases a month, her total utility from soda increases from 75 units to 123 units.
- For Lisa, the marginal utility from the second case each month is 48 units (123 – 75).
- The marginal utility numbers appear midway between the quantities of soda because it is the change in the quantity she buys from 1 to 2 cases that produces the marginal utility of 48 units.
- Marginal utility is positive, but it diminishes as the quantity of a good consumed increases.

Preferences

Total Utility and Marginal Utility



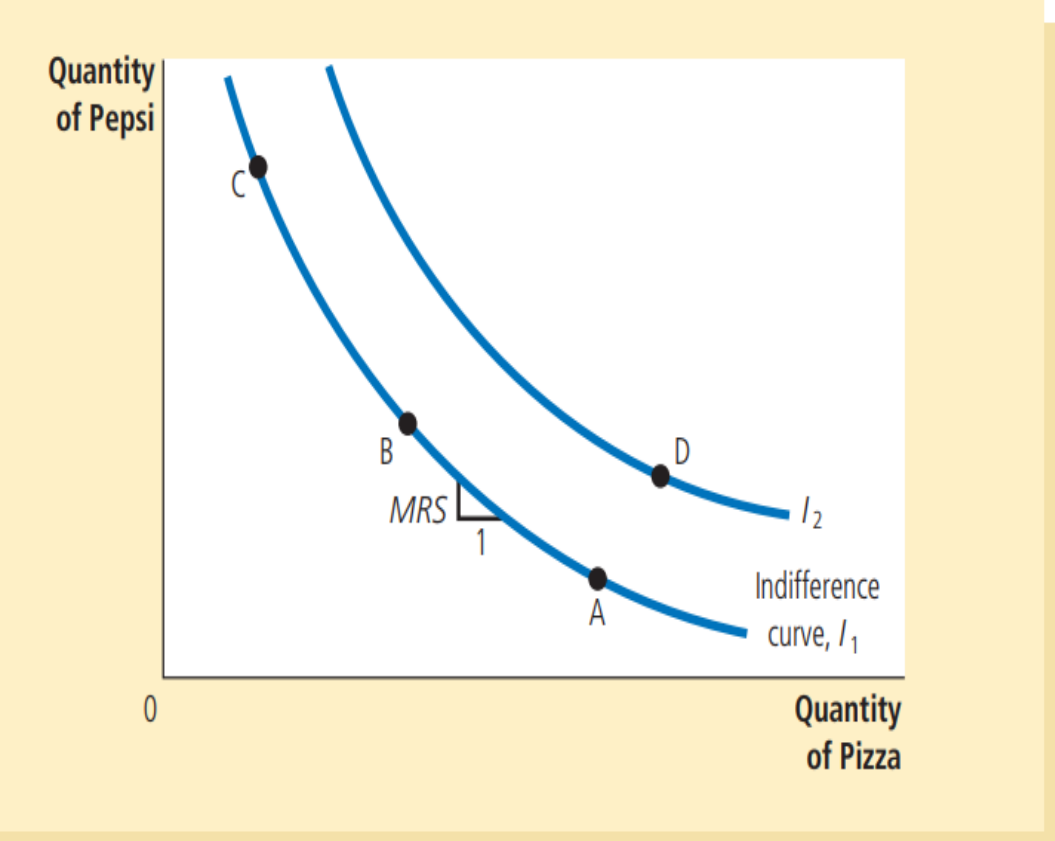
- Total utility in (a) rises with consumption, but it rises at a decreasing rate, showing diminishing marginal utility. This observation led early economists to formulate the law of downward sloping demand.
- The blue blocks show the extra utility added by each new unit. The fact that total utility increases at a decreasing rate is shown in (b) by the declining steps of marginal utility. If we make our units smaller, the steps in total utility are smoothed out and total utility becomes the smooth blue curve in (a). Moreover, smoothed marginal utility, shown in (b) by the blue downward-sloping smooth curve, becomes indistinguishable from the slope of the smooth curve in (a

Representing Preferences with Indifference Curves

Indifference curve - A curve that shows consumption bundles that give the consumer the same level of satisfaction.

The consumer's preferences allow her to choose among different bundles of pizza and Pepsi. If you offer the consumer two different bundles, she chooses the bundle that best suits her tastes. If the two bundles suit her tastes equally well, we say that the consumer is indifferent between the two bundles.

Representing Preferences with Indifference Curves



- This figure shows two of the consumer's many indifference curves.
- The consumer is indifferent among combinations A, B, and C because they are all on the same curve.
- Not surprisingly, if the consumer's consumption of pizza is reduced, say, from point A to point B, consumption of Pepsi must increase to keep her equally happy.
- If consumption of pizza is reduced again, from point B to point C, the amount of Pepsi consumed must increase yet again.

Representing Preferences with Indifference Curves

Marginal rate of substitution - the rate at which a consumer is willing to trade one good for another.

The slope at any point on an indifference curve equals the rate at which the consumer is willing to substitute one good for the other. This rate is called the marginal rate of substitution (MRS).

In this case, the marginal rate of substitution measures how much Pepsi the consumer requires to be compensated for a one-unit reduction in pizza consumption. Notice that because the indifference curves are not straight lines, the marginal rate of substitution is not the same at all points on a given indifference curve.

The rate at which a consumer is willing to trade one good for the other depends on the amount of the goods she is already consuming. That is, the rate at which a consumer is willing to trade pizza for Pepsi depends on whether she is hungrier or thirstier, which in turn depends on how much pizza and Pepsi she is consuming

Representing Preferences with Indifference Curves

Four Properties of Indifference Curves

- Property 1: Higher indifference curves are preferred to lower ones.
- Property 2: Indifference curves are downward sloping.
- Property 3: Indifference curves do not cross.
- Property 4: Indifference curves are bowed inward.